Simple Linear Regression – Marketing ROI

Project Overview

In this project, you will analyze a marketing dataset using Python and statsmodels to build a Simple Linear Regression model. You will learn how to prepare data, select predictive variables, implement OLS regression, validate model assumptions, and interpret statistical outputs to drive business decisions. This project helps you understand how data science translates raw data into actionable marketing insights.

Importing Necessary Libraries

```
%matplotlib inline
import pandas as pd
```

Loading the marketing data into a `DataFrame`

```
# Import Data Files from Google Drive


import requests
import pandas as pd
from io import StringIO
def read_gd(sharingurl):
    file_id = sharingurl.split('/')[-2]
    download_url='https://drive.google.com/uc?export=download&id=' + file_id
    url = requests.get(download_url).text
    csv_raw = StringIO(url)
    return csv_raw

url = "https://drive.google.com/file/d/10hWl9J084H7sx8LVJ1zzb47B0f9FdxGG/view?usp=sharing"
gdd = read_gd(url)

df = pd.read_csv(gdd)
```

```
df.head(-5)
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181
...	...	...	...	...
4562	54.0	26.077597	3.808179	188.723202
4563	93.0	25.285149	2.805840	327.466288
4564	99.0	36.024174	4.288755	355.807121
4565	19.0	1.490192	1.485684	66.393800
4566	16.0	2.339222	0.563121	52.423800

4567 rows × 4 columns

Next steps: Generate code with df New interactive sheet

Exploratory Data Analysis (EDA)

Data Overview

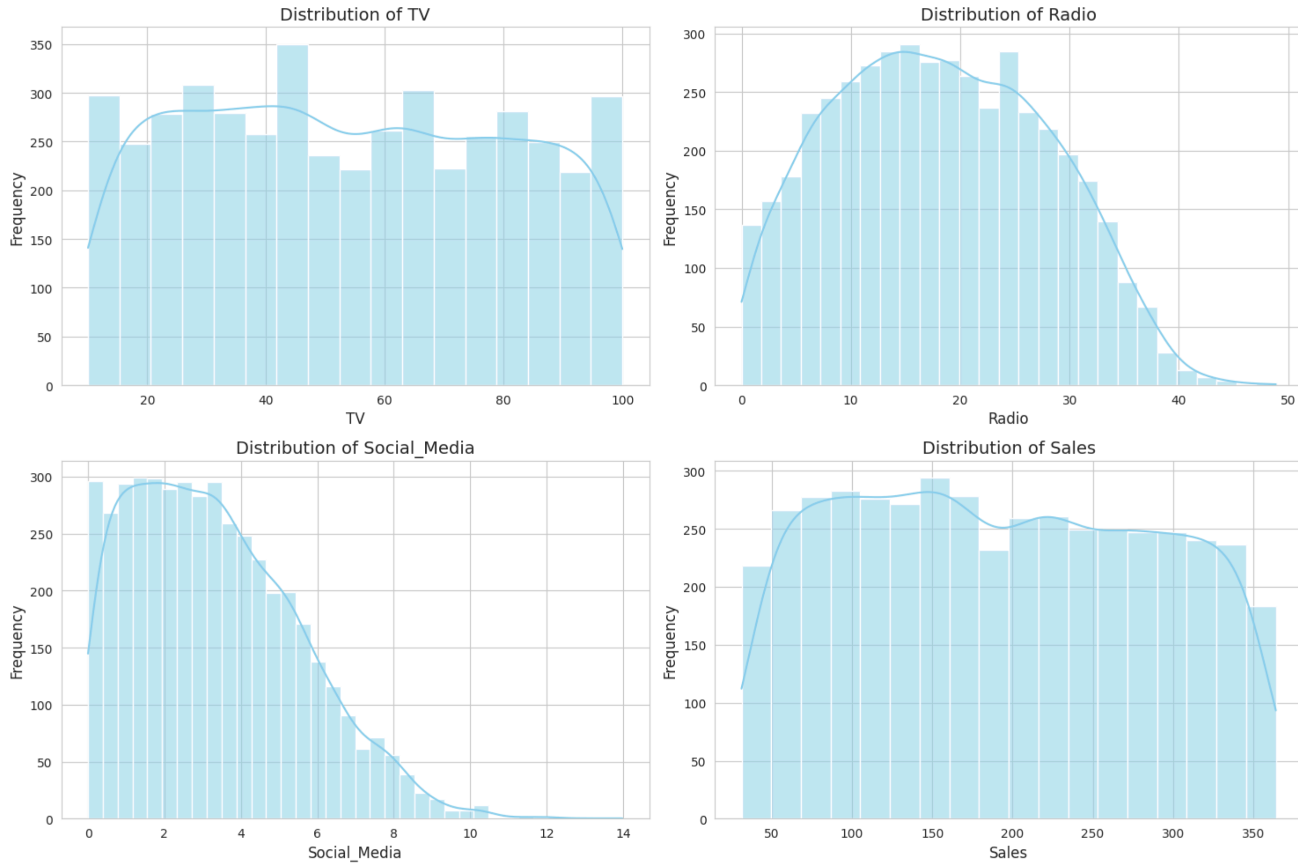
```
print(df.info())
display(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   TV            4562 non-null   float64
 1   Radio         4568 non-null   float64
 2   Social_Media  4566 non-null   float64
 3   Sales         4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
None
```

	TV	Radio	Social_Media	Sales
count	4562.000000	4568.000000	4566.000000	4566.000000
mean	54.066857	18.160356	3.323956	192.466602
std	26.125054	9.676958	2.212670	93.133092
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.525957	1.527849	112.322882
50%	53.000000	17.859513	3.055565	189.231172
75%	77.000000	25.649730	4.807558	272.507922
max	100.000000	48.871161	13.981662	364.079751

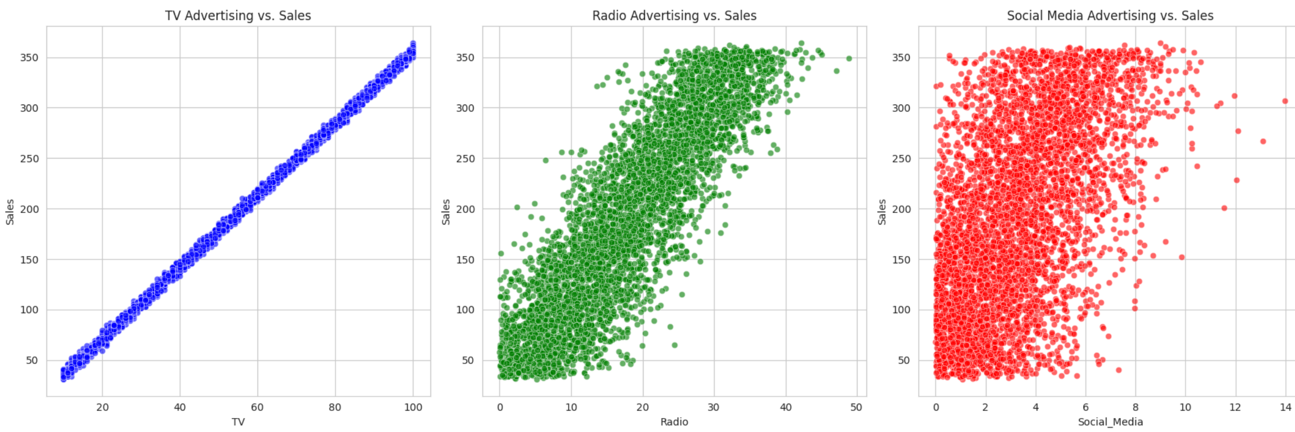
Distribution of Features

```
import matplotlib.pyplot as plt
import seaborn as sns

# Set a style for the plots
sns.set_style('whitegrid')

# Create a figure with subplots
fig, axes = plt.subplots(nrows=2, ncols=2, figsize=(15, 10))
axes = axes.flatten()

# Plot histogram and KDE for each numerical column
for i, col in enumerate(df.columns):
    sns.histplot(df[col], kde=True, ax=axes[i], color='skyblue')
    axes[i].set_title(f'Distribution of {col}', fontsize=14)
    axes[i].set_xlabel(col, fontsize=12)
    axes[i].set_ylabel('Frequency', fontsize=12)

plt.tight_layout()
plt.show()
```



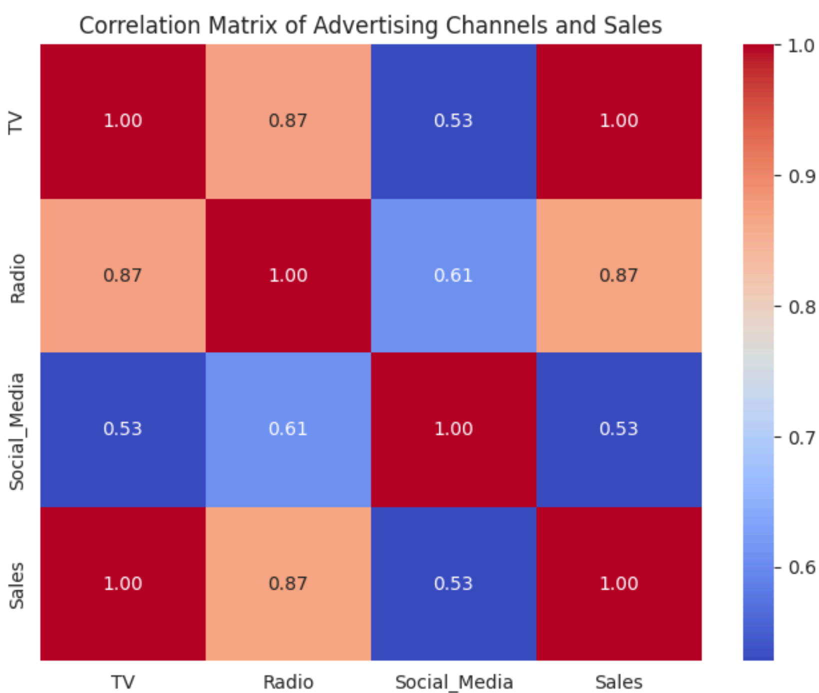
Relationship between Advertising Channels and Sales

```
fig, axes = plt.subplots(nrows=1, ncols=3, figsize=(18, 6))

sns.scatterplot(x='TV', y='Sales', data=df, ax=axes[0], color='blue', alpha=0.6)
axes[0].set_title('TV Advertising vs. Sales')

sns.scatterplot(x='Radio', y='Sales', data=df, ax=axes[1], color='green', alpha=0.6)
axes[1].set_title('Radio Advertising vs. Sales')

sns.scatterplot(x='Social_Media', y='Sales', data=df, ax=axes[2], color='red', alpha=0.6)
axes[2].set_title('Social Media Advertising vs. Sales')

plt.tight_layout()
plt.show()
```



Correlation Matrix

```
plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Matrix of Advertising Channels and Sales')
plt.show()
```



Handling Missing Values

```
# Check for missing values before imputation
print("Missing values before imputation:")
display(df.isnull().sum())

# Impute missing values with the mean of each column
for column in df.columns:
    if df[column].isnull().any():
        df[column] = df[column].fillna(df[column].mean())

# Check for missing values after imputation
print("\nMissing values after imputation:")
display(df.isnull().sum())
```

```
Missing values before imputation:
```

	0
TV	10
Radio	4
Social_Media	6
Sales	6

dtype: int64

```
Missing values after imputation:
```

	0
TV	0
Radio	0
Social_Media	0
Sales	0

dtype: int64

Building an OLS Regression Model with `statsmodels`

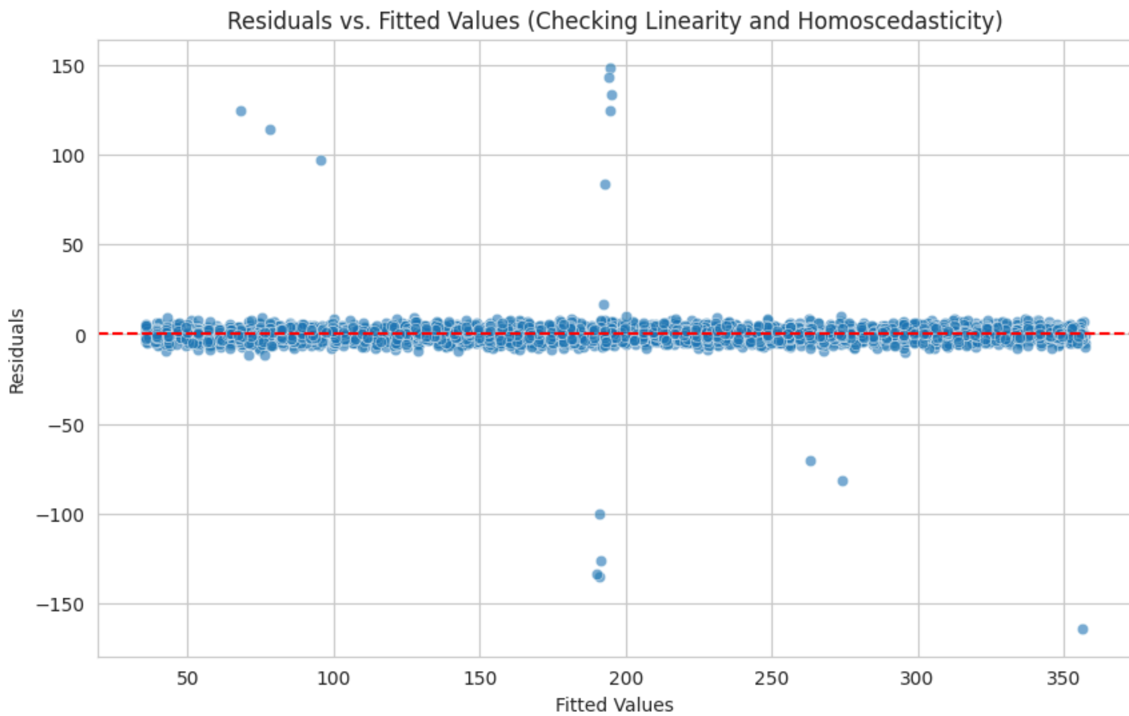
```
import statsmodels.formula.api as smf

# Define the regression formula
# Sales is the dependent variable, TV, Radio, and Social_Media are independent variables
model_formula = 'Sales ~ TV + Radio + Social_Media'

# Create and fit the OLS model
ols_model = smf.ols(formula=model_formula, data=df).fit()

# Print the model summary
print(ols_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.993
Model:                            OLS   Adj. R-squared:                  0.993
Method:                 Least Squares   F-statistic:                 2.279e+05
Date:                Fri, 12 Jun 2026   Prob (F-statistic):               0.00
Time:                        20:53:59   Log-Likelihood:                -15749.
No. Observations:                4572   AIC:                         3.151e+04
Df Residuals:                    4568   BIC:                         3.153e+04
Df Model:                           3
Covariance Type:            nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
Intercept        0.1372      0.264      0.520      0.603      -0.380       0.654
TV               3.5124      0.009    407.306      0.000       3.495       3.529
Radio            0.1234      0.025      4.962      0.000       0.075       0.172
Social_Media     0.0554      0.064      0.868      0.385      -0.070       0.181
==============================================================================
Omnibus:                     2763.181   Durbin-Watson:                   2.002
Prob(Omnibus):                  0.000   Jarque-Bera (JB):         11718659.357
Skew:                           1.133   Prob(JB):                         0.00
Kurtosis:                     251.012   Cond. No.                         149.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Diagnostic Plots for OLS Regression

```
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm

# Get residuals and fitted values from the OLS model
residuals = ols_model.resid
fitted_values = ols_model.fittedvalues
```

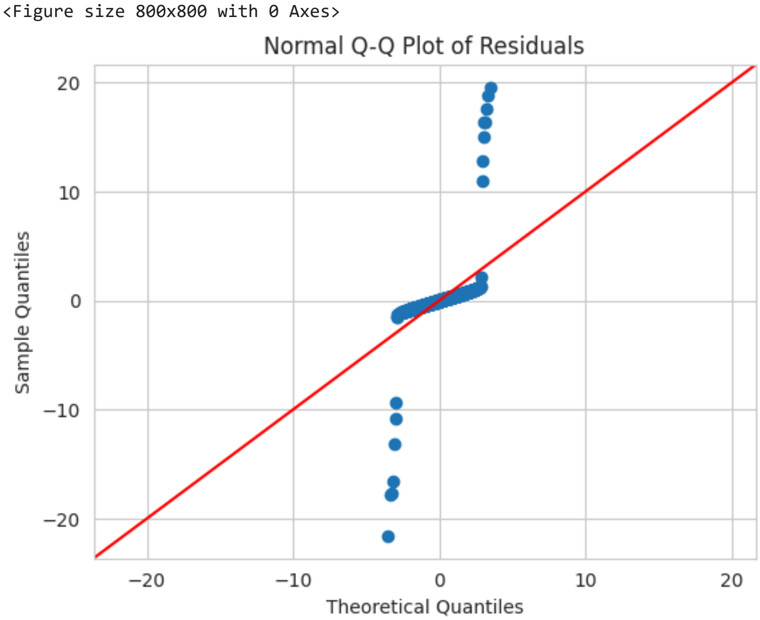
```
plt.figure(figsize=(10, 6))
sns.scatterplot(x=fitted_values, y=residuals, alpha=0.6)
plt.axhline(y=0, color='red', linestyle='--')
plt.title('Residuals vs. Fitted Values (Checking Linearity and Homoscedasticity)')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.grid(True)
plt.show()
```



Normality of Residuals (Q-Q Plot)

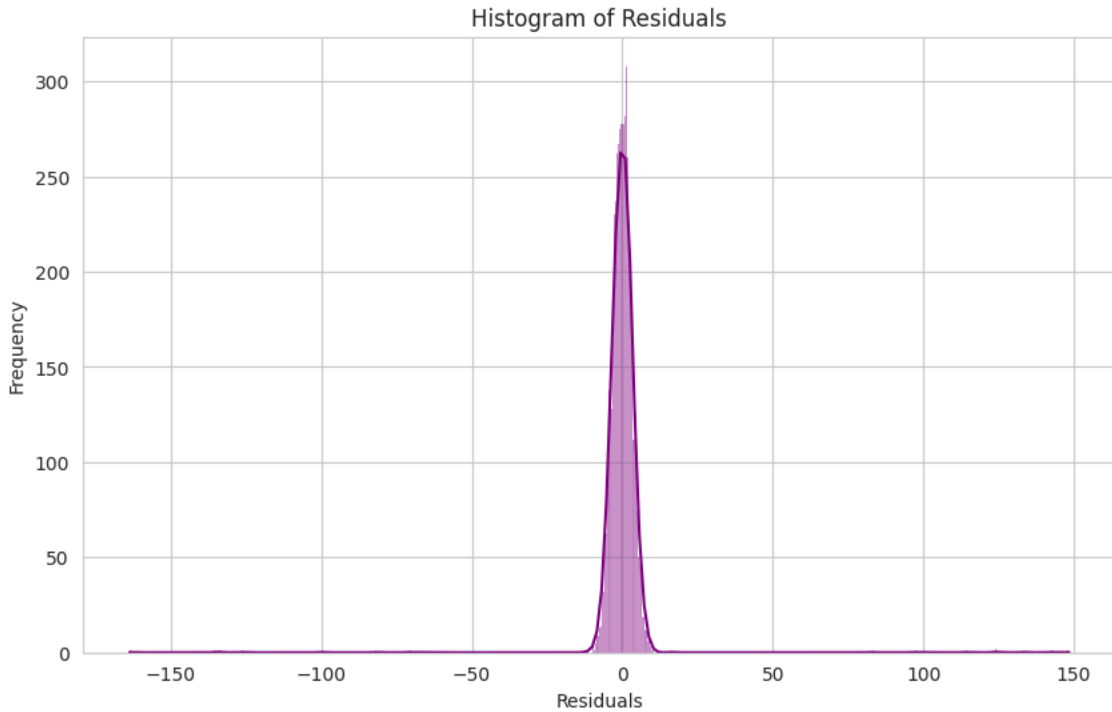
```
fig = plt.figure(figsize=(8, 8))
sm.qqplot(residuals, line='45', fit=True)
plt.title('Normal Q-Q Plot of Residuals')
plt.show()
```

<Figure size 800x800 with 0 Axes>



Normality of Residuals (Histogram)

```
plt.figure(figsize=(10, 6))
sns.histplot(residuals, kde=True, color='purple')
plt.title('Histogram of Residuals')
plt.xlabel('Residuals')
plt.ylabel('Frequency')
plt.grid(True)
plt.show()
```



Interpreting R-squared, coefficients, and p-values in business context

R-squared (0.993): Business Impact

An R-squared of 0.993 is exceptionally high, indicating that approximately 99.3% of the variation in Sales can be explained by the spending on TV, Radio, and Social Media advertising.

From a business perspective, this is fantastic! It suggests that our advertising efforts are highly effective in driving sales, and we have a very good understanding of how these efforts translate into revenue. It implies that these three channels are the dominant factors influencing sales, and there's very little 'unexplained' variability.

Coefficients: Understanding the Impact of Each Channel

TV (Coefficient: 3.5124): This is a highly significant and impactful coefficient. In a business context, it means that for every additional dollar (or unit) spent on TV advertising, Sales are predicted to increase by approximately $3.51.

This suggests that TV advertising is a very efficient and powerful channel for generating sales, providing a substantial return on investment.

Radio (Coefficient: 0.1234): This coefficient is also statistically significant, though smaller than TV's. It implies that for every additional dollar (or unit) spent on Radio advertising, Sales are predicted to increase by approximately $0.12. While not as high as TV, it still indicates a positive and measurable impact on sales, suggesting that radio advertising contributes to the overall sales performance.

Social_Media (Coefficient: 0.0554, not statistically significant): The positive coefficient suggests a potential increase in sales, but its lack of statistical significance (p-value of 0.385, much greater than 0.05) is crucial. In a business context, this means that, given the presence of TV and Radio advertising, we cannot confidently say that Social Media advertising has a statistically significant, independent impact on Sales. While it might contribute, its effect is too small or too noisy to be distinguished from zero within this model. This doesn't necessarily mean social media is ineffective overall, but rather that its direct and independent contribution to sales, when TV and Radio are also active, isn't clear from this model. It might be contributing in conjunction with other channels, or its impact might be harder to quantify directly with this linear model.

P-values: Confidence in Our Decisions

TV and Radio (p-values: 0.000): The extremely low p-values for TV and Radio advertising mean that there's a virtually zero chance that their observed positive impact on sales is due to random chance. This gives us high confidence in allocating resources to these channels and expecting a positive return. Social_Media (p-value: 0.385): The high p-value for Social Media advertising indicates that there's a 38.5% chance that the observed positive effect is just random noise. This means we cannot be confident that increased spending on Social Media alone will directly lead to a statistically significant increase in sales, if TV and Radio are already present. From a business decision-making standpoint, this suggests that simply increasing social media spend might not yield the expected direct sales lift, and a more nuanced strategy or further investigation might be needed for this channel.

Overall Business Implications:

Prioritize TV and Radio: The model strongly suggests that TV and Radio are highly effective advertising channels for driving sales. Businesses should consider maintaining or even increasing investment in these areas where appropriate.

Re-evaluate Social Media Strategy: The current model doesn't show a statistically significant direct impact from Social Media. This doesn't mean abandoning it, but rather investigating its role further.

High Predictability: The very high R-squared means that sales are highly predictable based on advertising spend, allowing for better forecasting and budget allocation for TV and Radio.

ROI-Based Marketing Budget Allocation Recommendation

ROI-Based Recommendation for Marketing Budget Allocation:

1. **Prioritize and Maximize Investment in TV Advertising:**
    - **Rationale:** TV advertising demonstrates the highest and most significant ROI. Every dollar spent yields the largest direct return in sales. This is your most efficient and impactful channel.
    - **Action:** Continue to allocate a substantial portion of the marketing budget to TV. Explore opportunities to optimize TV campaigns for even greater reach and impact, given its proven effectiveness.
2. **Maintain and Strategically Optimize Radio Advertising:**
    - **Rationale:** Radio advertising also shows a statistically significant positive return on investment, although lower than TV. It's a contributing factor to sales.
    - **Action:** Continue investing in Radio advertising. Consider optimizing ad placements, messaging, and target audiences to potentially enhance its ROI. It might serve as a valuable complementary channel to TV, reinforcing brand message and reaching